



Contents lists available at ScienceDirect

Small Ruminant Research

journal homepage: www.elsevier.com/locate/smallrumres

Association of *ASIP* and *MC1R* genotypes with the quantitative color variation in Huacaya alpaca

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ARTICLE INFO

Keywords:

Alpaca breeding
Fiber pigmentation
Haplotype
Color genotype
Gene interaction

ABSTRACT

The aim of this study was to identify the variants, genotypes, and haplotypes for the *ASIP* and *MC1R* genes in vicuñas and Huacaya alpacas, and to assess the association of these variants with fiber lightness and chromaticity values. The lightness and chromaticity values of the fiber for black/brown-black, dark brown/light brown, and white Huacaya alpacas, and vicuña were measured and grouped using the CIE L*a*b system. Genomic DNA was isolated from 89 alpacas and 9 vicuñas. Exons 2 and 4 of the *ASIP* gene and the coding sequence of *MC1R* were amplified and sequenced. Two non-synonymous variants (c.292 C>T and c.353 G>A) and a 57 bp in-frame deletion (c.325_381del) were identified in exon 4 of *ASIP*, and c.82 A>G, c.259 A>G, c.376 A>G, c.587 T > C, and c.901 C>T single nucleotide variants (SNV) plus a frameshift 4 bp deletion were identified in *MC1R*. Association analysis was carried out to relate the color phenotypes with variants and haplotypes, and genotypes of both genes. Loss-of-function recessive alleles (a¹, a², and a³) of *ASIP* gene were more frequently associated with black/brown-black fibers. All black/brown-black alpacas have at least one E allele of *MC1R* gene. Thus, an interaction between the recessive *ASIP* alleles with the dominant alleles of *MC1R* defines the black/brown-black alpaca color. In conclusion, alpaca fibers with the lowest lightness and lowest chromaticity tend to exhibit black/brown-black and dark brown phenotypes, although it was not possible to identify a clear cut-off between these groups. The a¹a¹/E-, a¹a²/E-, a³a³/E-, and a¹a³/E- genotypes and variants should be used as a selection marker for black/brown-black alpacas.

1. Introduction

The Peruvian Technical Standard 231.301 (Norma Técnica Peruana, 2022) classifies whole alpaca fleeces into 8 basic colors: white, light fawn X, light fawn Y, light fawn Z, light brown, dark brown, dark brown-black, and black, not including patches. In Australian alpacas, 8

phenotypes are also described based on fiber and skin pigmentation (Cransberg, 2017). On the other hand, at the industrial level, it is possible to make color combinations that would result in up to 16 natural colors (Australian Alpaca Association, 2012; Norma Técnica Peruana, 2022). However, the nomenclature used for brown phenotypes is ambiguous and subjective, depending on the visual

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<https://doi.org/10.1016/j.smallrumres.2026.107704>

Received 9 June 2025; Received in revised form 27 December 2025; Accepted 7 January 2026

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interpretation of the evaluator and/or each producer. Therefore, the use of objective color measurements based on standard colorimetry assays would be very useful.

In the recent past, in the local, national, and international markets, black and brown fiber had a lower commercial value than white fiber (Oria et al., 2009; Pinares et al., 2023) because of its lower demand by the textile industry compared to white fiber. However, this trend is currently being reversed because of a focus on the more environmentally friendly natural colors (Cruz et al., 2021). From the point of view of sustainability, it is crucial first to increase, and then stabilize the population of naturally colored alpacas.

To select for color uniformity, the assignment of color phenotype descriptions based on objective measures, such as quantitative colorimetry, should be implemented (Pinares et al., 2021; Cruz et al., 2021). For example, the CIE $L^*a^*b^*$ system aims to replicate human perception of color and would therefore be a good tool to use to measure the color of alpaca fiber. The L^* value indicates lightness, and the a^* and b^* values represent chromaticity on the green-red and blue-yellow scale, respectively. This standardized measurement has already been widely used to analyze alpaca fiber color (Lupton et al., 2006; Guridi et al., 2011; Gray et al., 2023; Lozano et al., 2023, 2024; Baxter et al., 2025).

Color in fiber-producing animals has been one of the most important selection traits (Cieslak et al. 2011; Linderholm and Larson, 2013; Librado et al., 2017). The alpaca was domesticated from the vicuña (Gentry et al., 2004; Fan et al., 2020) and has undergone continuous selection over centuries for different fiber colors (e.g. white, shades of brown and black), which has led to introgression of *de novo* variants in color genes into the population (Marín et al., 2018). The *ASIP* gene (agouti signaling protein) and the *MC1R* gene (melanocortin 1 receptor), both are the main regulators of the synthesis and proportional distribution of eumelanin and pheomelanin in the fiber and skin of many species, including alpacas (Feeley and Munyard, 2009; Feeley et al., 2011; Cransberg, 2017; Marín et al., 2018), although more than 150 genes associated with color expression have been documented in other animals (e.g. Cieslak et al., 2011; Rochus et al., 2019, reviewed in Caro and Mallarino, 2020).

The single nucleotide variants and haplotypes in the *ASIP* and *MC1R* genes in colored alpacas and llamas, and the association of these variants with white, light brown, dark brown and black fiber has been described (Powell et al., 2008; Feeley and Munyard, 2009; Feeley et al., 2011; Guridi et al. 2011; Chandramohan et al., 2013, 2015, Daverio et al., 2016, Pallotti et al., 2020). Studies by Chandramohan et al. (2015), Cransberg (2017), and Anello et al. (2022) have provided insights into the skin transcriptomes of colored alpacas and llamas. Cransberg et al. (2013), Cransberg (2017), and Wakamatsu et al. (2021) also characterized and quantified melanin types in alpacas. Together, these studies advanced our understanding of the genetic basis of fleece color determination in alpacas and llamas.

Nevertheless, the molecular mechanisms of pigmentation remain poorly understood, particularly the ability to reliably distinguish between the “brown” color groups. Therefore, the objective of this study was to identify the variants, genotypes, and haplotypes for *ASIP* and *MC1R* genes in vicuñas and colored Huacaya alpacas, and to assess their association of these variants with lightness and chromaticity values.

2. Materials and methods

2.1. Fiber color measurement

Raw fiber samples were collected from the mid-side of 14 vicuñas (*Vicugna vicugna mensalis*). Nine of the vicuñas were from the Centro Ecológico Recreacional Taraccasa in the Provincial Municipality of Abancay (Peru), and five of the fleeces were from the Textile Fiber Laboratory, Universidad Nacional Agraria La Molina in Lima, Peru. Collection of the vicuña samples was authorized by the Servicio Nacional Forestal y de Fauna Silvestre - SERFOR (RDG N° 270-2019-

MINAGRI-SERFOR-DGGSPPFFS). Additionally, 89 raw samples of Huacaya alpaca fiber (*Vicugna pacos*) were collected, and were sorted using the Pacamarca color chart (Cruz et al., 2021), including eight white (WH), 24 light brown (LB), 19 dark brown (DB), 11 dark brown-black (DBL), and 27 black (BL). Examples of these colors are shown in Fig. 1.

The fiber samples were manually cleaned by removing visual impurities, then scoured using a standardized procedure. These samples were dried and measured under a standard environment of temperature ($20 \pm 2^\circ\text{C}$) and relative humidity ($65 \pm 5\%$).

Previously, Pinares et al. (2021) and Cruz et al. (2021) described and validated the sample handling and color measurement protocol using the same colorimeter device on alpacas. The color of the cleaned samples was quantified using a Chroma Meter CR-210 (Konica Minolta, USA). This device was calibrated using a standard illuminant D65 based on the procedure of the Commission Internationale de l'Eclairage (Schanda, 2007).

According to the CIE $L^*a^*b^*$ system, L^* values range from 0 (black) to 100 (white). The a^* axis indicates the green-red scale. Values for a^* ranging from -100 to $+100$ indicate green to red scale, respectively. The b^* axis represents blue-yellow scale, values range from -100 to $+100$, indicating blue to yellow, respectively (Druml et al., 2018; Ly et al., 2020). CIE $L^*a^*b^*$ values were recorded using the technique of holding the wadded sample against the machine's aperture so that no light could penetrate the sample. The distance from the central axis for the a^* and b^* values represent chromaticity (C^*), or a color's saturation and this was estimated based on the formula described by Druml et al. (2018) and Ly et al. (2020): $C^* = \sqrt{(a^*)^2 + (b^*)^2}$.

The normality of the colorimetry data was tested using the Shapiro-Wilk normality test, and the lightness and chromaticity values did not conform to a normal distribution. Therefore, differences within and between all the color groups, including alpaca and vicuña samples, on lightness and chromaticity were evaluated by Kruskal-Wallis and statistical significance determined using Dunn's test ($\alpha = 0.05$). Since no significant differences were found between black and brown-black, the 5 initial alpaca color groups were reduced to 4 groups (i.e. black/brown-black, dark brown, light brown, and white) for the haplotype association analysis.

The color groups were ranked from darkest to lightest, then Spearman's rank correlation was carried out to assess the relationship between the color groups with lightness and chromaticity values, as described by Nakashima et al. (2022).

2.2. Extraction of DNA and amplification of genes

Blood was collected from the same 89 alpacas and the 9 vicuñas that were sampled for fiber color. Genomic DNA was extracted from 200 μL of whole blood using the Quick-DNA™ Miniprep Plus Kit (Zymo Research, UK). Throughout the study, genomic DNA and amplicon concentration and quality was assessed via spectroscopy at 260 nm and 260/280 ratio using a NanoDrop™ 8000 (Thermo Fisher Scientific, UK), and DNA integrity was confirmed using agarose gel electrophoresis.

For all polymerase chain reactions (PCR) ~ 75 ng of genomic DNA was used as template, and the reactions were completed in a Mastercycler® Nexus - PCR Thermal Cycler (Eppendorf, USA). Details of primers, reagents, expected amplicon size and cycling conditions are provided in Supplementary Table 1. Exon 2 of the *ASIP* gene (VicPac4 chr19 GCF_048564905.1 NC_133005.1:25550258–25627818) was amplified from 17 of the darkest color alpaca DNA samples using the protocol and primers described by Daverio et al. (2016). *ASIP* exon 4 was amplified from 89 alpacas and 9 vicuñas DNA samples, also using the primers from Daverio et al. (2016). The 5' flanking region of exon 2 included part of the intron 1 sequence, and the 3' flanking region of exon 4 included part of the 3' UTR sequence.

Similarly, the *MC1R* coding sequence (GCF_048564905.1 NC_133007.1:33194817–33209702; note that the *MC1R* gene is not



Fig. 1. Alpaca phenotypic color variation according to the Pacamarca color chart. From left to right: light brown, dark brown, white, black, and dark brown-black.

correctly annotated in the VicPac4 assembly, and currently is encompassed within exon 2 of the *Tubulin beta-3 chain* gene), including the entire single-exon coding region and flanking regions, was amplified from the same 89 alpacas and 9 vicuñas using the PCR protocol and primers described by Pinares et al. (2021) and Daverio et al. (2016) respectively.

All primers were tested *in silico* using the reference genome, VicPac4 (GCF_048564905.1) to verify the desired amplification locations on *ASIP* and *MC1R*, because they had been designed using earlier versions of the alpaca genome. The size of the amplicons was confirmed by electrophoresis on a 1 % agarose gel containing Hi-SYBr Safe Gel Stain (Himedia HiGenoMB®, India) and a 3000 bp molecular size marker.

2.3. Sequencing, analysis of variants and haplotypes

The three independent amplicons were purified using the USB ExoSAP-IT® enzyme according to the manufacturer's recommendations (Thermo Fisher Scientific, USA) and were sequenced using dideoxynucleotide cycle sequencing (Sanger sequencing) in both forward and reverse directions using the original PCR primers (Macrogen Inc., Seoul, South Korea). The quality of the forward and reverse sequences was checked according to the percentage of high quality (HQ), then alignment was carried out by ClustalW (Thompson et al., 1994) using Geneious software version 11.1.5.

Single nucleotide variants (SNV) in both genes were identified using the consensus from the nine vicuña sequences (generated in this study) as a reference. In addition, we compared our sequencing data with the *ASIP* and *MC1R* alpaca GenBank sequences accession numbers EU220010, HM768324, HM768323, HQ645017, and HQ645018 (Feeley and Munyard, 2009; Feeley et al., 2011; Chandramohan et al., 2013, 2015). For this, we used the BLAST nucleotide alignment tool from NCBI, achieving a similarity of > 99 %. Non-synonymous variants were identified based on the coding sequence of both *ASIP* and *MC1R* genes, from the amino acid sequence predicted using the translation function of Geneious software version 11.1.5. Based on the non-synonymous variants in the *ASIP* and *MC1R* amplicons, haplotypes were determined using the Bayesian ELB algorithm of Harlequin software version 3.5.2.2, taking into account the gametic phase > 0.9 (Excoffier and Lischer, 2010).

2.4. Association analysis of fiber color with *ASIP* and *MC1R* genes

For each non-synonymous variant, an association analysis between the genotypes and lightness and chromaticity was independently carried out using Kruskal-Wallis and significance was determined using Dunn's

test ($\alpha = 0.05$). Spearman's rank correlation was also used to analyze the relationship of the *ASIP* and *MC1R* genotypes, based on each independent variant, with the lightness and chromaticity values, respectively. The genotypes were coded according to the number of specific alleles present (i.e. 0/1/2).

The associations of the haplotypes of *ASIP* and *MC1R* genes with the color groups were independently assessed using Fisher's exact test ($\alpha = 0.05$). As described before, non-synonymous variants were used to identify the haplotypes of *ASIP* and *MC1R*, respectively. Four and eight haplotypes were identified for exon 4 of *ASIP* (2 SNV and 1 deletion of 57 bp) and *MC1R* (5 SNV), respectively. The *ASIP* and *MC1R* haplotypes were used to align the denomination of alleles with the nomenclature described by Feeley et al. (2011), plus the wild allele (vicuña). All statistical analyses were performed using RStudio on the R software (R Core Team, 2022).

2.5. Predicting the effects of variants on protein functionality

The probability of potential damage (Pdel) on the functionality of *ASIP* and *MC1R* proteins caused by single nucleotide variants was determined using the PANTHER (Protein Analysis Through Evolutionary Relationships) and PANTHER-PSEP (position-specific evolutionary preservation) programs, as described by Tang and Thomas (2016). The effect of the deletion on *MC1R* was approximated using MutationTaster (<https://www.mutationtaster.org/>) with the human *MC1R* transcript ENST00000555147; which has GCTT in place of the deleted ACTT in alpacas, is 85 % similar at the DNA level and 89 % by amino acid.

3. Results

3.1. Phenotypes based on fiber colorimetry

The measurement of lightness and chromaticity of the fiber samples in the alpaca color groups showed lower lightness and chromaticity values in black and brown-black alpacas (Table 1). The five alpaca color groups were reduced to three classical colors as a result of the Dunn test and 95 % confidence intervals. The two darkest groups (black/brown-black) and the two brown groups (dark brown/light brown) were collapsed. See Supplementary Table 2 for individual colorimetric values.

The relationship of the rank of the phenotypically assigned color groups with objective lightness and chromaticity values (assessed by Spearman's rank correlation) was strongly positive and significant ($p < 0.0001$). Lightness (correlation = 0.85) and chromaticity (correlation = 0.92) both decrease from white, vicuña, light brown, dark

Table 1

Evaluation of lightness and chromaticity values on vicuña and colored Huacaya alpaca fiber.

#	Color groups *	Lightness (L*)		Chromaticity (C*)	
		Median	CI _s	Median	CI _s
14	Vicuña	47.73 ^b	45.6–48.9	24.40 ^a	23.5–24.6
08	White	84.58 ^a	83–87.6	08.71 ^c	08–9.1
24	Light brown	28.90 ^c	27.5–30	16.65 ^b	15–17.2
19	Dark brown	25.99 ^c	24.4–26.9	13.55 ^b	12.3–15
11	Dark brown-black	18.44 ^d	17.9–18.6	01.91 ^d	1.8–2.1
27	Black	17.85 ^d	17.6–18	01.03 ^d	0.9–1.1

Number of animals by color. * Alpaca phenotypes originally assigned according to the Pacamarca alpaca color chart. Different letters in the same column indicate significantly different medians (Dunn, $\alpha = 0.05$). CI_s: The 95 % confidence intervals for the medians.

brown, to black/brown-black.

3.2. Single nucleotide variants and haplotypes of the ASIP gene associated with phenotypes

The ASIP exon 2 sequence was studied in only 17 darker-colored alpacas out of 89, in order to confirm previously described variants and check for new ones. No non-synonymous variants were found in exon 2 (Supplementary Table 3).

In 89 alpacas and 9 vicuñas, the exon 4 ASIP sequence was sequenced. In alpacas, 3 variants were found in exon 4; 2 non-synonymous SNV and a deletion of 57 bp. The c.292 C>T (p.R98C), c.353 G>A (p.R118H), and c.325_381del (p.C109_R127del); Table 2). No variants were found in the ASIP exon 4 sequence in vicuñas. For most variants, the alternative allele was significantly associated with lower median lightness and chromaticity (Table 2)

Lightness and chromaticity values were also significantly associated with the ASIP genotypes for the three SNV based on the Spearman rank correlation ($p < 0.05$). The c.292 C>T SNV, has a low probability of altering protein function ($P_{del} = 0.27$). However, the probability of protein damage from c.353 G>A is moderate ($P_{del} = 0.57$).

Four haplotypes were identified in exon 4 of ASIP, named H1 (TGN), H2 (C-D), H3 (CAN), and H4 (CGN) (Fig. 2). All vicuñas had the H4 haplotype, although this haplotype was present in all color groups.

Haplotypes H1, H2, and H3 were significantly associated with black/brown-black alpaca fleece ($p < 0.001$), while H4 was associated with some white and light brown alpacas and with all vicuñas ($p < 0.005$). Furthermore, when alpaca phenotypes were grouped into two categories: black/brown-black versus brown (i.e. dark and light), a significant association of H3 with black and H4 with brown was also observed ($p < 0.001$), although 50 % of brown alpacas have H3 (Fig. 2).

The four previously described alleles of the ASIP gene (Feeley et al.,

2011) were matched to the haplotypes ($a^1=H2$, $a^2=H1$, and $a^3=H3$). However, for the first time, H4 was able to be designated as the wild allele (A^+) because it was found in a homozygous state in 100 % of the vicuñas. Black/brown-black alpacas were homozygous for, or compound heterozygous for one of the three recessive alleles (a^1 , a^2 , or a^3). None of the other alpaca color groups were homozygous for any of the recessive alleles (Table 3).

3.3. Single nucleotide variants and haplotypes of the MC1R gene associated with phenotypes

The MC1R exon was amplified and sequenced in 87 alpacas and 9 vicuñas. The coding sequence was 100 % identical in vicuñas. Compared to vicuñas, 5 non-synonymous SNV were found in alpacas (c.82 A>G, c.259 A>G, c.376 A>G, c.587 T > C, and c.901 C>T). In addition, a deletion (c.224_227del) was found only in alpacas. This deletion was observed in black (26.32 %), brown (19.51 %), and white animals (25 %), but no animal was homozygous for this variant. The alternative allele of all non-synonymous MC1R SNV was significantly associated with higher median lightness and chromaticity (Table 4). In addition, these positive associations for the different SNV with fiber lightness and chromaticity were confirmed by Spearman's correlation analysis ($p < 0.001$). We explored the additive and dominance effects on lightness values using a linear multiple regression and found them only for SNV c.82 A>G and c.224_227del. Thus, 82 AG was always additive when 224_227del was additive or dominant. Similarly, 82AG was always dominant when 224_227del was additive or dominant.

Eight haplotypes were obtained from the five non-synonymous SNV: H1: AAATC, H2: AGGTC, H3: GGGTT, H4: GGGTC, H5: GAGTT, H6: GGGCC, H7: GGATT, and H8: AGGTT (Fig. 3). H1 was associated with the black/brown-black and with the light and dark brown color groups ($p < 0.001$). H4 was associated with alpaca of light brown and white colors, as well as vicuña ($p < 0.001$). H6 was only associated with light brown ($p < 0.01$).

Haplotype H1 (aka E), whether homozygous or heterozygous, was present in 100 % of black/brown-black alpacas, 84 % of dark brown, and 86 % of light brown alpacas, and was not observed in white alpacas. Black alpacas showed a higher frequency of the H1 haplotype in heterozygosity (0.63) than in homozygosity (0.37), so H1 is the dominant allele E. H4 was homozygous in 100 % of the vicuñas, so it is likely to be the wild allele E^+ ; it was also present in 62.5 % of white alpacas and 18.18 % of light brown alpacas. Since H6 was observed in only 8 out of 22 light brown alpacas and in 2 out of 11 brown-black alpacas, it may be the recessive allele e.

The coding sequence (CDS) of MC1R encodes a functional protein of 317 amino acids, whereas the CDS c.224_227del allele encodes a truncated protein of 84 amino acids. This change was predicted to be a loss of function variant (high confidence) due to the loss of multiple

Table 2

Evaluation of fiber lightness and chromaticity in relation to SNV genotypes of the ASIP gene in alpacas and vicuñas.

SNV	Genotype	n	Lightness			Chromaticity		
			Median	SE	Spearman(r_s)	Median	SE	Spearman(r_s)
c.292 C>T (p.R98C)	CC	77	26.72 ^a	2.31	0.37 ($p < 0.000$)	11.80 ^a	0.90	0.32 ($p < 0.003$)
	CT	17	18.44 ^b	0.86		1.56 ^b	1.49	
	TT	02	17.45 ^b	0.54		1.04 ^b	0.01	
c.325_381de (p.C109-R127)	NN	64	26.82 ^a	2.34	0.22 ($p < 0.04$)	13.89 ^a	1.01	0.35 ($p < 0.001$)
	ND ₅₇	30	19.85 ^a	3.60		2.02 ^b	1.01	
	D ₅₇ D ₅₇	02	19.19 ^a	2.20		0.80 ^b	0.12	
	AA	16	18.20 ^b	0.99		1.50 ^b	1.40	
	GA	26	25.85 ^{ab}	2.61		15.05 ^a	1.23	
c.353 G>A (p.R118H)	GG	22	44.11 ^a	4.80	0.45 ($p < 0.000$)	17.05 ^a	1.81	0.47 ($p < 0.000$)
	GD	16	20.29 ^b	6.49		6.61 ^b	1.40	
	AD	14	19.85 ^b	1.17		1.89 ^b	1.43	
	DD	02	19.19 ^b	2.19		0.80 ^b	0.12	

Different letters in columns indicate significant differences in medians (Dunn, $\alpha = 0.05$). Spearman's rho (r_s) with p-values. D₅₇ = deletion of 57 bp, N = No deletion sequence, D = variant not present due to presence of D₅₇ deletion. Note: 96 out of the 98 genotyped animals were used for this analysis.

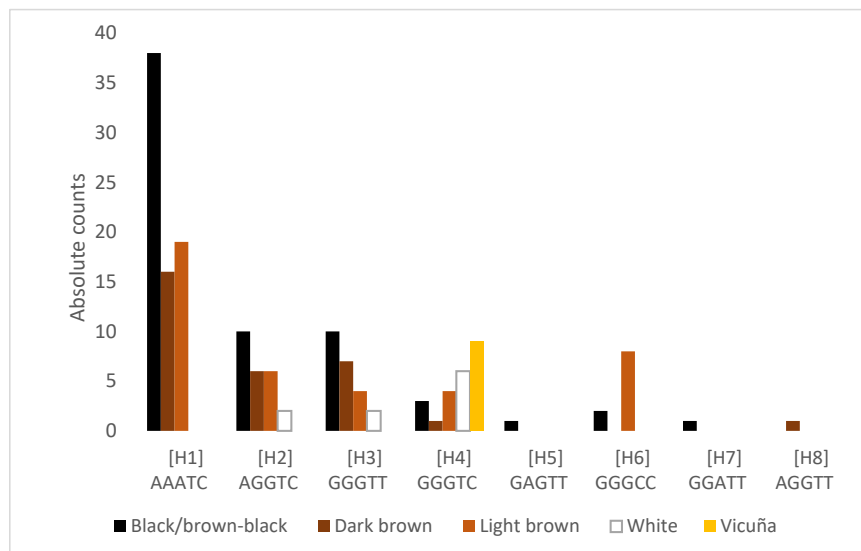


Fig. 3. Frequency of *MC1R* SNV haplotypes in alpaca and vicuña.

Table 5

Interaction of the *ASIP* and *MC1R* alleles and their resulting color phenotypes.

<i>ASIP</i> and <i>MC1R</i> genotypes	Colorimetry-based color groups
a^1a^1/E^-	Black/brown-black
a^2a^3/E^-	Black/brown-black
a^2a^2/E^-	Black/brown-black
a^1a^3/E^-	Black/brown-black
a^3a^3/EE	Black/brown-black
a^1a^2/E^-	Black/brown-black
A^+a^3/E^-	Dark/light brown
a^3-/E^-	Dark/light brown
A^+a^3/E^+	Dark/light brown
A^+A^+/E^+	White
A^+a^1/E^+E^+	White
A^+A^+/E^+E^+	Vicuña

Wild-type allele in the *ASIP* gene: A^+ and recessive alleles: a^1 , a^2 , and a^3 . In the *MC1R* gene, the wild allele (E^+), dominant (E), and the recessive (e).

observed due to the homozygous presence of the wild allele (A^+/E^+) of both genes (Table 5).

4. Discussion

Our results confirmed a strong correlation between variants of *ASIP* and *MC1R* with lightness and chromaticity values. These results support the use of colorimetry as a powerful tool for color assignment, which helps breeders to select for uniform darker color (Baxter et al., 2025). Lightness measures the degree of whiteness of the fibers, the median in white Huacaya alpaca was 85.21. This value is comparable with 87.79 ± 1.53 and 87.14 ± 1.43 (mean $\pm$ SD) reported by Lozano et al. (2023, 2024), and with 87.10 ± 3.97 reported by Cruz et al. (2021). However, it is higher than reported by Guridi et al. (2011), 73.58.

Slight variations in yellowness or lightness in white alpacas may be influenced by environmental factors, such as a physical light-scattering phenomenon (Baxter et al., 2025). These variations may also be related to fiber diameter and uniformity. In our study, these variations could also be due to the small sample size of white alpacas ($n = 8$), as well as to the disproportionate number of samples in each color category. Thus, fibers with high lightness would have a low amount, or absence, of eumelanin; conversely in dark brown and black fibers, a higher amount of eumelanin corresponds to lower lightness values (Cecchi et al., 2011; Cransberg et al., 2013).

The white fibers had the highest lightness, and lightness values

decreases from light to dark fibers. For instance, light brown and black fibers had medians of 28.90 and 17.85, respectively. These values are like those reported by Cruz et al. (2021), Guridi et al. (2011) and Gray et al. (2023). In this regard, Cecchi et al. (2011) and Cransberg et al. (2013) indicated that there is a predominant presence of eumelanin in black alpaca, with little or minimal presence of eumelanin in brown alpaca, where pheomelanin is more abundant. In addition, brown color in alpacas may be related to the variation of melanin content, where the eumelanin/pheomelanin ratio is low (Cransberg et al., 2013). Dilution of eumelanin may be due to defective tyrosinase synthesis pathways, which are encoded by the *TYR* gene and other genes, such as tyrosinase-related proteins 1 and 2 (Gray et al., 2023).

Among alpacas, white and brown colors predominate in herds, although there are discrepancies among evaluators in the assignment of fleece color by visual appraisal (Cruz et al., 2021). Therefore, the use of a standardized color measurement will avoid bias in color assignment (Cruz et al., 2021; Baxter et al., 2025). Black and brown-black assigned by visual appraisal did not show differences in lightness and chromaticity (Table 1), so they were grouped together as a black color. This finding aligns with those reported by Cruz et al. (2021) and Gray et al. (2023).

The environment affects the color intensity of alpaca fiber as sun exposure degrades eumelanin and pheomelanin, resulting in a dilution of color at the tip of the fiber (Wakamatsu et al., 2021). In Peru's high-altitude regions, exposure to intense sunlight and frost can bleach the tips of alpaca fiber, which can sometimes affect color measurements. This phenomenon may affect the brown-black alpacas, even though they have the same genotype as black. It was hypothesized that these differences could be due to a higher production of eumelanin in black than brown-black alpacas, with no difference in pheomelanin production (Wakamatsu et al., 2021).

Although colorimetry helps to objectively assign colors in alpacas (Lupton et al., 2006; Cruz et al., 2021), another alternative method, such as pigment quantification (eumelanin versus pheomelanin) could be used to assign the genotype to the color phenotype (Fan et al., 2010; Cecchi et al., 2011; Cransberg et al., 2013; Wakamatsu and Ito, 2021). The absence of a chemical analysis of eumelanin and pheomelanin content in our study may limit the reliability of phenotype as an indicator of genotype. However, this method would be too costly and time-consuming for Peruvian alpaca breeders. Therefore, this study is the first to perform a fiber color assignment based on lightness and chromaticity using the CIE $L^*a^*b^*$ system and the association of *ASIP* and *MC1R* variants and haplotypes. These findings will facilitate the

selection of alpacas with darker pigmentation based on reliable genotypes, thereby increasing the production of colored fiber for the textile industry.

The predicted lengths of *ASIP* exon 2 (160 bp) and exon 4 (177 or 120 bp) were confirmed by sequencing. These lengths and exon annotations are consistent with those of the alpaca reference genome VicPac4 (https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_048564905).

1/). In *ASIP* exon 2, only one novel SNV c.102 G>A (p.G34G) was found, but the non-synonymous polymorphisms: c.11 C>G (p.T4S) and c.19 T>C (p.F7L) reported by Almathen et al. (2018) and Chandramohan et al. (2013) were also found. On the other hand, the two SNV reported in exon 2 of *ASIP* in *Camelus dromedarius* (Almathen et al., 2018), a 1 bp deletion (c.23T_{del}) and a non-synonymous SNV (c.25 G>A), with two alleles, c.23T_{del}/25 A in black and dark brown and c.23 T/c.25 G in light brown and white camels were not found in our study. Exon 3 of *ASIP* was not sequenced because Feeley et al. (2011), Chandramohan et al. (2013), Daverio et al. (2016), and Marín et al. (2018) did not report any non-synonymous SNV in that region.

In vicuñas, exon 4 of *ASIP* contained no variants because it is a wild specie that has not undergone artificial selection for uniform fleece color, which is consistent with the report by Marín et al. (2018). In alpacas, we found two contiguous SNV, one synonymous and one non-synonymous, called double nucleotide polymorphisms (DNP), also reported by Feeley et al. 2011; Chandramohan et al. 2013; and Marín et al. 2018. Although they were not analyzed as such in these studies, they will be taken into account in future analyses as they may have an impact on the structure and function of the protein or may even provide information on the genetic diversity of the species.

The heterozygous deletion (D₅₇/-) was present in 12 black and six brown-black alpacas (0.45), and the D₅₇/D₅₇ genotype was observed at low frequency, 2 out of 29 black alpacas (0.07). Feeley et al. (2011) observed D₅₇/D₅₇ in 21 of 50 black alpacas (0.42) and the D₅₇/- in 19 of 50 (0.35). In contrast, Chandramohan et al. (2013) did not report the presence of D₅₇/D₅₇ genotype. They observed the D₅₇/- genotype in 17 alpacas and found no deletion in 10 out of 27 black alpacas. These frequency differences undoubtedly reflect the gene pool in different populations.

Alpacas with eumelanin pigmentation presented at least one of the three non-synonymous variants: c.292 C>T (T/T), c.353 G>A (A/A), and c.325_381del (D₅₇/D₅₇). The combination of two variants as compound heterozygotes was also observed: the deletion (D₅₇/-) with C/T, as well as C/T with G/A. In this regard, Chandramohan et al. (2013) and Feeley et al. (2011) have proposed genotypes based on the three recessive alleles of the *ASIP* gene, indicating that black alpacas present the three variants in homozygosity, the deleted one, c.292 T (p.R98C) and c.353 A (p.R118H), or in heterozygosity the combination of two of these variants.

The haplotypes H1 and H2 of *ASIP* were found most frequently in black/brown-black alpacas, haplotype H3 was found in all colors with the exception of white alpacas, while H4 was found mainly in non-black alpacas (Fig. 2). Haplotypes H1 and H4 differ in c.292 C>T, although the probability of altering protein function is low (P_{del} = 0.27). Haplotypes H3 and H4 differ by c.353 G>A, the homozygous A variant could be associated with the production of eumelanin and GG with pheomelanin. For this variant, the probability of protein damage is moderate (P_{del} = 0.57). The haplotypes H1, H2 and H3 were matched with the previously reported alleles with loss of function as a¹ = H2, a² = H1, and a³ = H3, in agreement with Feeley et al. (2011) and Chandramohan et al. (2013), and H4 was assigned as the wild allele (A⁺), because it was present homozygously in 100 % of vicuñas. However, all of the vicuñas were from the same region, so this finding would need to be validated in other populations.

On the other hand, of the five non-synonymous variants in the *MC1R* gene only the c.224_227del and c.587 T>C variant were not associated with lightness and chromaticity. In addition, the c.259 A>G and c.376 A>G variants showed a moderate and positive correlation with

the lightness and chromaticity values (p < 0.0001); the variant c.82 A>G showed a low and positive correlation with the lightness and chromaticity values (p < 0.01) (Table 4). Guridi et al. (2011) also reported associations of three SNV (c.82 A>G, c.865 C>T, and c.901 C>T) with fiber lightness (p < 0.05), suggesting them as important markers for selection, although we did not found variants at c.865 C>T. In white alpacas from the United States, Australia, and Peru, the T/T and C/T genotypes have been reported in c.901 C>T (Powell et al., 2008; Feeley and Munyard, 2009; Chandramohan et al., 2015), but in our study, the small sample size of white alpacas would limit to ability to show a strong association with the c.901 T variant.

Other authors reported non-synonymous variants that were not found in our study, such as: c.383 C>T (Powell et al., 2008); c.865 C>T, c.874 G>A, and c.886 T>C (Guridi et al., 2011); c.92 C>T (Chandramohan et al., 2015); and c.265. A>G (Marín et al., 2018).

The *MC1R* deletion c.224_227del was also reported by Powell et al. (2008); similar to our results, they found it at heterozygosity in 1 of 5 dark browns, 2 of 11 medium browns, 1 of 9 light browns, 5 of 16 dark creams, 2 of 9 light fawns (LF), and 3 of 7 whites. Also, Chandramohan et al. (2015) found the c.224_227del deletion in black, brown, and white alpacas. However, this deletion has not been found in other studies of alpacas and llamas (Feeley and Munyard, 2009; Guridi et al., 2011; Marín et al., 2018; Daverio et al., 2016). The absence or presence of this variant in alpacas is probably due to the selection pressure to which they have been subjected and the frequency of the allele in the founding population.

Feeley and Munyard (2009) and Chandramohan et al. (2015) identified E and e alleles for the *MC1R* gene. Taking into account the haplotypes with significant association, H1, H4, and H6 may be the alleles E, E +, and e, respectively. The AAATC/AAATC (EE) genotype may be associated with black and brown-black, AAATC/GGGCC (Ee) with light brown, and GGGTC/GGGTC (E⁺E⁺) with vicuña (p < 0.001). The AAATC/AAATC (EE) genotype is in alignment with the AAAC/AAAC genotype described by Chandramohan et al. (2015) and may also be equivalent to the haplotype consisting of c.82 G>A and c.901 C>T and the AC/AC genotype associated with black and brown-black alpacas (Feeley and Munyard, 2009). Previous studies in Arabian dromedary camels have shown that the EE genotype is associated with dark phenotypes (black and dark brown), whereas the Ee and ee genotypes are associated with light brown and white (Almathen et al., 2018).

More than 36.84 % of dark brown alpacas showed the EE genotype, and the rest maintained the heterozygous genotype (E/-), in agreement with the report of Shang et al. (2019). Light brown alpacas showed greater allelic variability in the *MC1R* gene, where 18.18 % had the wild allele in homozygosity, 36.36 % with a heterozygous genotype AAATC/GGGCC (E/e), and only 13.64 % with a dominant genotype EE. Possibly the ee genotype is present in lighter colors such as light fawn or white. In contrast, Chandramohan et al. (2015) reported only a heterozygous genotype (Ee) for light brown, while Feeley and Munyard (2009) reported the EE, Ee, and ee genotypes for light fawn, but neither author has considered the wild allele of vicuña.

An epistatic mode of action among the alleles of the *MC1R* and *ASIP* genes may occur in most of black alpacas. The dominant E and recessive e alleles of the *MC1R* gene occur in combination with the recessive alleles of the *ASIP* gene (a¹, a², and a³) in homozygosity and/or heterozygosity, which by segregation would theoretically result in 12 possible genotypes, consistent with those proposed by Hart et al. (2003), Feeley et al. (2011), Edens and Gill (2022). However, Pallotti et al. (2020) reported that the E allele in the homozygous state only interacts with the recessive alleles of *ASIP* in the heterozygous state (alpacas from Quimsachata, Peru).

In general, the exclusive presence of eumelanin pigment in black fiber may depend on the presence of the a¹ allele (57 bp deletion), and the brown-black fiber would be related to the presence of the a³ allele (Feeley et al., 2011; Edens and Gill, 2022). Therefore, not all black and brown alpacas in our study exhibited interaction because they were

heterozygous for the *MC1R* and *ASIP* genes, as reported by Pallotti et al. (2020). However, our study had one limitation. The absence of pedigree data prevented us from conducting a segregation analysis of the epistatic interaction between the *ASIP* and *MC1R* genes. This analysis would improve the accuracy with which we can predict fiber color in future generations.

In most dark brown alpacas and some light brown and white alpacas, the dominant allele (E) is observed in combination with the wild allele A^+ in heterozygosity (A^+a^3), resulting in 4 possible genotypes, consistent with the report by Pallotti et al. (2020). On the other hand, in black and brown-black horses, the genotypes have an epistatic interaction of aa/EE, aa/Ee or Aa/EE, while in brown horses the genotypes are AA/EE and AA/Ee (Druml et al., 2018; Shang et al., 2019).

The conservation of homozygous wild alleles at the *MC1R* and *ASIP* genes in vicuñas confirms their status as the ancestral color of alpacas, as described by Marín et al. (2018). The segregation of true black fleece is very complex due to the epistatic interactions between *ASIP* and *MC1R* alleles, also future research will have to take in account the effect of other genes, such as tyrosinase-related protein 1 and 2, the membrane-associated transporter protein (*MATP*), and the receptor tyrosine kinases (*KIT*). To obtain a black alpaca, a pair of recessive *ASIP* alleles may be required, so mating black animals with different genotypes (e.g., $a^1a^2 \times a^2a^3$, $a^1a^3 \times a^1a^2$, etc.) could be used to maintain the frequency of alleles in the herds (Anello et al., 2022). Colorimetry defined by lower lightness and lower chromaticity will be very useful to select black and dark-brown animals with genotypes a^3a^3/EE , $a^2a^2/E-$, or $a^1a^3/E-$ (Feeley et al., 2011; Pallotti et al., 2020; and Edens and Gill, 2022).

Because many genes may influence the pigment intensity or dilution in the alpaca fiber, an alternative method for identifying novel mutations in alpacas is a genome-wide association study (GWAS) complemented with long-read sequencing. Successfully identifying a marker (s) for darker pigment intensity will allow breeders to select black and brown fleece colors more accurately (Gray et al., 2023).

5. Conclusions

At the molecular level, the wild alleles of *ASIP* and *MC1R* genes were identified in vicuñas and alpacas. An association of single nucleotide variants and haplotypes of *ASIP* and *MC1R* genes with lightness and chromaticity values in alpaca fibers ranged from low to moderate. Black/brown-black Huacaya alpacas have the E allele in homozygosity and/or heterozygosity, combined with the recessive a^1 , a^2 , and a^3 alleles. Dark brown, light brown, and white alpacas may exhibit minimal epistatic interaction between *ASIP* and *MC1R* genes due to the expression of the A^+ and E^+ wild alleles. However, no interaction has been observed in vicuñas due to the conservation of the wild alleles.

CRedit authorship contribution statement

Kylie A. Munyard: Writing – review & editing, Visualization, Formal analysis, Methodology, Validation. **María Silvana Daverio:** Writing – review & editing, Supervision, Methodology. **Rubén Pinares:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Gustavo A. Gutiérrez-Reynoso:** Writing – review & editing, Visualization, Supervision. **Alan Cruz:** Visualization, Writing – review & editing.

Ethical/legal aspects

The collection of biological samples was carried out in compliance with the standards of animal protection and welfare (Law No. 30407), as well as what is indicated in RDG No. 270–2019-MINAGRI-SERFOR-DGGSPFFS.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Acknowledgements

The authors acknowledge the financial support from the National Council for Science, Technology and Technological Innovation (CONCYTEC), PhD Scholarships in Animal Science through Grant Agreement 178–2015-FONDECYT. We thank the University of Natural Resources and Life Sciences (BOKU) for the loan of the colorimeter.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.smallrumres.2026.107704.

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